

Problem Statement

The tokenization of African languages remains a significant challenge in the development of large language models (LLMs). Most existing tokenizers are trained on datasets dominated by high-resource languages, causing languages such as Igbo, Yoruba, and Swahili to be fragmented into excessive tokens. This increases computational costs, reduces context efficiency, and can negatively affect translation quality, text generation, and other downstream language tasks. Although several initiatives have expanded datasets for African languages, comparatively little attention has been given to optimizing how these languages are tokenized. This project extends SuperBPE to better capture the linguistic characteristics of Igbo, Yoruba, and Swahili, while introducing **EncodeLang**, a community-driven platform that enables native speakers to contribute validated translations for evaluating tokenizer performance. The project aims to improve language accessibility, reduce inference costs, and support more inclusive AI systems for African communities.